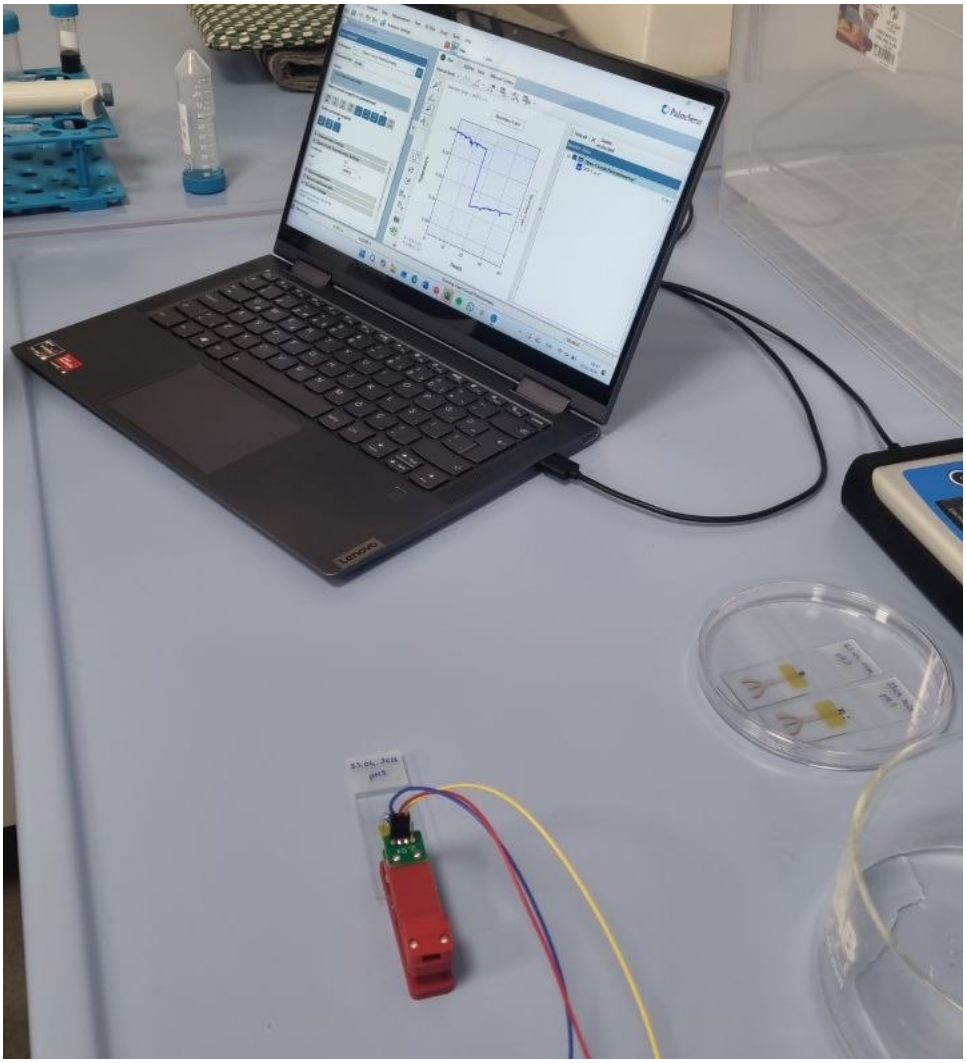
**24.04.2026**

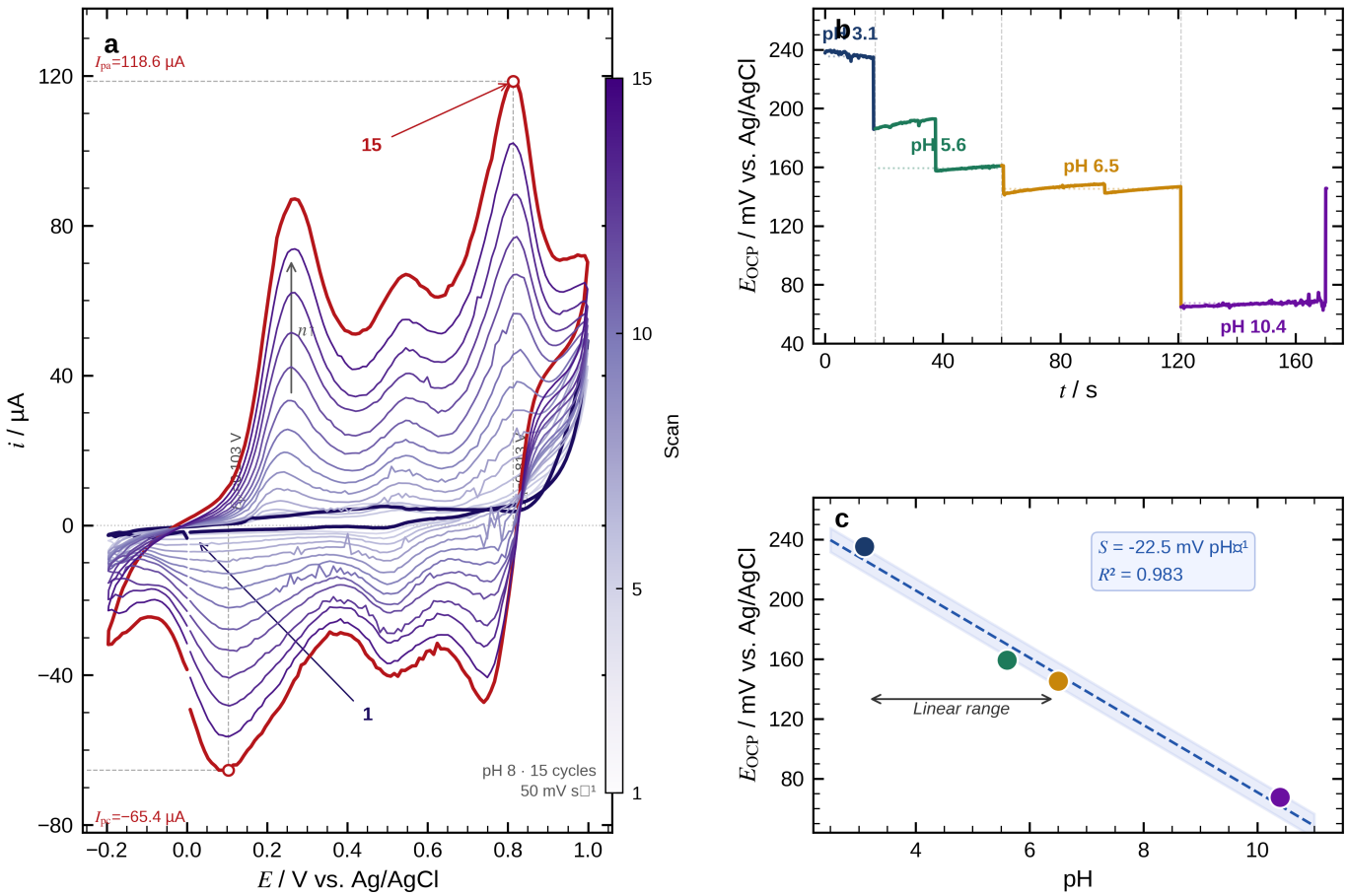
## Report on the first characterization results of the pH sensor:

All electrochemical measurements were performed using a PalmSens4 potentiostat connected to a custom screen-printed electrode (SPE). The SPE consists of three electrodes printed on a glass substrate: a carbon working electrode, a carbon counter electrode, and a silver reference electrode (Ag). Measurements were carried out under static drop-casting conditions, where small volumes of buffer solution (30 µL) were pipetted directly onto the electrode surface.



**Sensor Preparation:** Aniline Electropolymerizing. To make the electrode sensitive to pH, the working electrode surface was functionalized with polyaniline (PANI), a conducting polymer whose electrical properties change in response to the acidity of the surrounding solution. This was done by placing the

electrode in a solution of aniline monomer (0.1 M in 1 M HCl) and applying a cycling voltage between −0.2 V and +1.0 V vs. Ag/AgCl for 15 consecutive cycles at a scan rate of 50 mV $s^{-1}$, a process called cyclic voltammetry (CV). During this process, aniline molecules in the solution are oxidized at the electrode surface and progressively build up a thin PANI polymer film directly on the working electrode.




CV Results : Confirming Film Formation. The CV data confirmed that the PANI film was successfully deposited and became increasingly electroactive with each cycle. The peak current associated with PANI oxidation grew from 46.0 µA in scan 1 to 118.6 µA in scan 15, a 2.6-fold increase, indicating that more and more electroactive polymer was accumulating on the surface. Simultaneously, the reduction peak deepened from −3.4 µA to −65.4 µA, showing that the film's redox process was becoming more reversible and the film was maturing. By scan 15, the film had reached a stable, reproducible electroactive state, confirming it was ready for pH sensing.

pH Sensing Open Circuit Potential Measurements. Once the PANI film was formed, the sensor's response to different pH levels was tested using Open Circuit Potential (OCP) measurement, a technique that simply measures the voltage the sensor naturally produces in contact with a solution, without applying any external current. A fresh drop of buffer at each pH was placed on the sensor, and the stable voltage was recorded after the signal settled.

The sensor was tested across five pH levels in ascending order: pH 3.1, 5.6, 6.5, 7.6, and 10.4. As pH increased, the OCP decreased, from 235 mV at pH 3.1 down to 68 mV at pH 10.4. This is physically expected: at low pH (acidic), the PANI film is fully protonated and highly conductive, producing a high voltage signal. As the solution becomes more basic, the film progressively loses protons and its conductivity decreases, lowering the voltage.

Calibration and Sensitivity. Plotting the stable OCP values against pH produced a calibration curve with a sensitivity of −21.1 mV per pH unit ($R^2 = 0.90$). This means that for every one unit increase in pH, the sensor voltage drops by approximately 21 mV, a clear and measurable signal. The large potential drop observed between pH 7.6 and 10.4 is attributed to the wide pH gap between these two measurement points (2.8 units). Additional measurements at closer pH intervals across the pH 8–10 range are planned to fully map the sensor response in the alkaline region.